

Technology News

Intelligent t-shirts to monitor patients

Scientists at the Carlos III University of Madrid have developed an intelligent t-shirt for patients that monitors the wearer's physiological parameters (temperature, heart rate, etc) in a non-intrusive manner and also locates him within the hospital. It can even determine whether the patient is seated, lying down, walking or running. The information gathered by the t-shirt using e-textile technology is wirelessly sent to an information management system, which then shows the patient's location and vital signs in real time.

The system consists of the fixed infrastructure, which is pre-installed in the hospital, and the mobile unit, which moves with the patient. The mobile unit includes the intelligent t-shirt and a localisation device, which can be carried in a pocket. The t-shirt is washable and includes electrodes that detect bioelectric power through which an electrocardiogram can be obtained. In addition, it has a removable device that includes a thermometer and an accelerometer, which are used to take the patient's temperature, relative position (reclining, standing, etc) and the level of physical activity.

Finally, the indoor localisation unit is activated periodically, receives signals from the units that make up the fixed localisation infrastructure and wirelessly sends that information to the information management system. Once the information is received there, the localisation algorithm is able to establish the individual's position within a two-metre margin of error, and to mark the spot on a map of the hospital.

LED lights to transmit data wirelessly

Till now LEDs were used only for illuminating. But now Harald Haas, a



Intelligent t-shirt that monitors the wearer's physiological parameters

professor of engineering at Edinburgh University, has discovered how LEDs can serve as wireless Internet access points. Car headlamps, street lamps or any other lighting device that uses LEDs can be used for communication by a new networking technology named 'Li-Fi.'

Hass's D-Light system uses a mathematical trick called orthogonal frequency-division multiplexing, which allows it to vary the intensity of the LED's output at a very fast rate, invisible to the human eye. The signal can be picked up by simple receivers.

As of now, the system has achieved data rates of up to 10 Mbps (faster than a typical broadband connection), which is expected to reach 100 Mbps by the end of this year.

Bionic leg

A new lower-limb prosthetic developed at Vanderbilt University allows amputees to walk without the leg-dragging gait characteristic of conventional artificial legs.

The device uses the latest advances

in computer, sensor, electric motor and battery technology to give it bionic capabilities. The prosthetic has powered



Professor Michael Goldfarb, right, with amputee Craig Hutto who is wearing the new bionic leg developed at Vanderbilt University